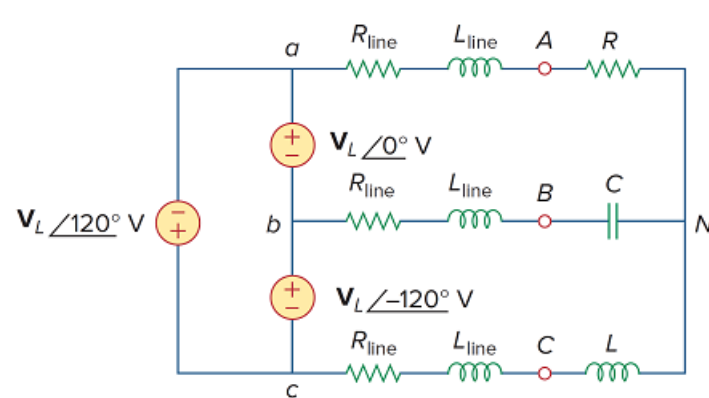


Problem

Using Fig. 12.68, design a problem to help other students better understand how to use *PSpice* or *MultiSim* to analyze three-phase circuits.

Figure 12.68



Step-by-step solution

Step 1 of 1

Using Fig. 12.68, design a problem to help other students better understand how to use *PSpice* to analyze three-phase circuits.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

The circuit in Fig. 12.68 operates at 60 Hz. Use *PSpice* to find the source current **lab** and the line current **lbB**.

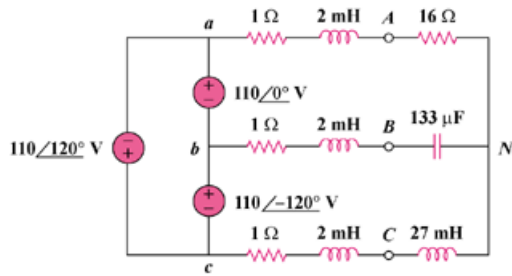


Figure 12.68

Solution

Because of the delta-connected source involved, we follow Example 12.12. In the AC Sweep box, we type Total Pts = 1, Start Freq = 60, and End Freq = 60. After simulation, the output file includes

```
FREQ      IM(V_PRINT2)  IP(V_PRINT2)
6.000 E+01  5.960 E+00   -9.141 E+01
FREQ      IM(V_PRINT1)  IP(V_PRINT1)
6.000 E+01  7.333 E+07    1.200 E+02
```

From which

lab = $7.333 \times 10^7 \angle 120^\circ$ A, **lbB** = $5.96 \angle -91.41^\circ$ A

